Reference Frame Discrepancy in TA Fluorescence Detector Survey Marker Coordinates

Impact on Opt-copter Calibration
and Implications for $X_{\max}$ Reconstruction

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1 Executive Summary

All eight original survey markers used for fluorescence detector (FD) telescope geometry at the three Telescope Array FD stations—Black Rock Mesa (BRM), Long Ridge (LR), and Middle Drum (MD)—are stored in the analysis database using NAD 83(CORS96)(Epoch:2002.0000) ellipsoidal heights. The Opt-copter RTK-GPS reports positions in WGS 84, which is operationally equivalent to the International Terrestrial Reference Frame (ITRF). The ellipsoidal height difference between these two frames at the TA site is approximately **0.73 m**, with NAD 83 heights being systematically higher than ITRF.

This discrepancy produces a systematic elevation angle bias of approximately $+0.17°$ in the Opt-copter pointing direction analysis at a flight distance of 250 m. This bias is uniform across all telescopes at each station and matches the observed ΔElevation offsets reported by Matsuzawa (2025) for Middle Drum and by Nakazawa (2020) for Black Rock Mesa. The effect propagates into shower geometry reconstruction: preliminary estimates suggest a systematic underestimation of $X_{\max}$ on the order of 9–15 g/cm^2 per 0.1° of pointing error, with corresponding energy biases of $\sim$4%.

The corrected ITRF coordinates are derived from the same OPUS static solutions that produced the original NAD 83 values. No resurvey is required for the seven well-determined markers. One marker (CLF1) must be resurveyed due to a failed OPUS solution.

2 Background

2.1 The Survey Campaign

In March 2011, a GPS survey campaign was conducted at all three TA FD stations to establish precise reference markers for the Opt-copter calibration system. A Trimble TRM41249.00 antenna was used for all occupations. Data were submitted to the NGS Online Positioning User Service (OPUS) for static processing. Each OPUS solution provides coordinates in two reference frames simultaneously:

- NAD 83(CORS96)(Epoch:2002.0000) — the North American datum
- ITRF00 (at the observation epoch) — the global reference frame

The NAD 83 coordinates were entered into the TA analysis database and used for all subsequent telescope geometry computations, including the Opt-copter direction analysis.

2.2 The Reference Frame Problem

NAD 83 and ITRF are distinct reference frames. NAD 83 is fixed to the North American tectonic plate, while ITRF is an Earth-centered, Earth-fixed global frame that accounts for plate motion. At any given location, the two frames differ by a combination of rotation, translation, and scale parameters that vary with position and epoch. At the TA site ($\sim$39.3° N, $\sim$113° W), the dominant manifestation is a systematic difference in ellipsoidal height of approximately 0.73 m, with smaller horizontal offsets of a few centimeters.

The Opt-copter's onboard RTK-GPS receiver reports positions in WGS 84. For all practical purposes, WGS 84 and ITRF are identical at the centimeter level. When the analysis code computes the geometric relationship between the Opt-copter (in WGS 84/ITRF) and the survey markers (stored in NAD 83), the height discrepancy introduces a systematic error in the computed elevation angle.

3 OPUS Solutions and Height Discrepancies

3.1 Original Markers

Table 1 presents the ellipsoidal heights from both reference frames as reported in the OPUS solutions, along with the height difference $\Delta h = h_{\mathrm{ITRF}} - h_{\mathrm{NAD83}}$. Where multiple OPUS solutions exist for a single marker (from the comprehensive dataset in the survey archive), inverse-variance weighted averages are used.

Table 1: Ellipsoidal height comparison: NAD 83 vs. ITRF for original survey markers. Heights are from OPUS static solutions (March 2011). The ITRF values represent weighted averages where multiple solutions exist.

Marker	Site	h_{NAD83} (m)	h_{ITRF00} (m)	σ_h (m)	Δh (m)	n
BR1	Black Rock	1395.791	1395.053	0.004	−0.738	7
BR2	Black Rock	1395.405	1394.659	0.009	−0.746	2
CLF2	Central Laser	1370.716	1369.979	0.016	−0.737	2
LR1	Long Ridge	1538.577	1537.847	0.009	−0.730	3
LR2	Long Ridge	1539.073	1538.340	0.010	−0.733	2
MD1	Middle Drum	1587.964	1587.222	0.004	−0.742	3
MD2	Middle Drum	1583.179	1582.449	0.020	−0.730	1
				Weighted mean:	−0.736	

The height offsets are remarkably consistent, ranging from −0.730 to −0.746 m, with a weighted mean of −0.736 m. The smooth latitude-dependent variation is expected from the frame transformation geometry.

3.2 Rejected Solution: CLF1

The OPUS solution for marker CLF1 at the Central Laser Facility is unusable:

- Observations used: 500 / 19,489 (3%)
- Fixed ambiguities: 18 / 26 (69%)
- Overall RMS: 0.118 m
- Peak-to-peak vertical uncertainty: 130.347 m

The CLF1 marker must be resurveyed. A short RTK occupation with EarthScope NTRIP corrections would provide a definitive ITRF position.

3.3 Mixed Reference Frames in the Current Database

An additional complication exists in the current marker database (Table 2). Newer markers surveyed after the original 2011 campaign use a different NAD 83 realization:

Table 2: Reference frame realizations in the current marker database.

Markers	Reference Frame	Notes
BR1, BR2, CLF1, CLF2, LR1, LR2, MD1, MD2	NAD 83(CORS96) (Epoch:2002.0000)	Original 2011 survey
BRNE, BRNW, BRSE, BRSW, MD3, MD4, MDNE, MDNW, MDSE, MDSW	NAD 83(2011) (Epoch:2010.0000)	Later survey

These two NAD 83 realizations differ by 1–2 cm in position, so the existing database is not even internally self-consistent, although this secondary effect is much smaller than the 0.73 m NAD 83-to-ITRF discrepancy.

4 Impact on Opt-copter Direction Analysis

4.1 Geometry of the Error

The Opt-copter flies at a distance $d \approx 250\,\mathrm{m}$ from the FD mirror center (MD measurements, Matsuzawa 2025) or $d \approx 300\,\mathrm{m}$ (BRM measurements, Nakazawa 2020). The monument coordinates define the reference geometry from which the Opt-copter's angular position is computed. If the monument height is biased by Δh, the computed elevation angle ε to the Opt-copter is systematically offset by:

$$\delta\varepsilon \approx \arctan\left(\frac{|\Delta h|}{d}\right) \tag{1}$$

For the MD station (using marker MD2, $\Delta h = -0.730\,\mathrm{m}$, $d = 250\,\mathrm{m}$):

$$\delta\varepsilon = \arctan\left(\frac{0.730}{250}\right) = 0.167^\circ \tag{2}$$

Because the NAD 83 heights are higher than ITRF, the monument appears elevated relative to the Opt-copter's true frame, causing the Opt-copter to appear systematically *lower* than expected. The direction analysis computes $\Delta\mathrm{Elevation} = \mathrm{GPS} - \mathrm{CoG}$, so a monument that is too high produces a *positive* ΔElevation offset: the GPS-derived position appears above the FD-measured center of gravity.

4.2 Comparison with Observed Offsets

4.2.1 Middle Drum (Matsuzawa 2025)

The Opt-copter measurements at MD (September–October 2024) show a systematic positive elevation offset across all 14 telescopes:

Table 3: MD telescope pointing offsets from Matsuzawa (TA-ALL, June 2025). Upper row: even-numbered FDs; lower row: odd-numbered FDs.

FD ID	02	04	06	08	10	12	14
ΔAz (°)	−0.02	+0.01	+0.01	+0.03	−0.02	±0.00	−0.10
ΔEl (°)	+0.15	+0.17	+0.14	+0.19	+0.20	+0.21	+0.25
FD ID	01	03	05	07	09	11	13
ΔAz (°)	+0.03	+0.10	+0.06	+0.04	+0.07	−0.01	+0.03
ΔEl (°)	+0.14	+0.07	+0.11	+0.19	+0.18	+0.22	+0.23

Key observations:

- **All 14 telescopes show positive ΔElevation**, ranging from +0.07° to +0.25°.
- The mean ΔElevation across all telescopes is approximately +0.17°.
- The azimuth offsets scatter around zero with no systematic bias.
- The predicted offset from the height discrepancy is $\arctan(0.730/250) = 0.167°$.

The agreement between the predicted 0.167° and the observed mean of ∼0.17° is striking.

4.2.2 Black Rock Mesa (Nakazawa 2020)

The BRM Opt-copter measurements (2018–2019) show a different pattern: a systematic *negative* elevation trend that increases with azimuth (telescope number). This is consistent with a combination of the height offset and an azimuthal pointing misalignment. The BRM results use markers BR1/BR2, which have $\Delta h \approx -0.74\,\mathrm{m}$, predicting $\delta\varepsilon = \arctan(0.74/300) = 0.141°$ at the 300 m flight distance used at BRM.

The BRM elevation offsets range from +0.11° (FD 00) to −0.19° (FD 09), with a pattern suggesting real telescope-to-telescope pointing variations superimposed on the systematic height-induced bias.

4.3 Discrimination from Other Error Sources

The uniform, positive ΔElevation signature at MD is the key diagnostic. Other potential error sources would produce different signatures:

Table 4: Expected signatures of different systematic error sources.

Error Source	ΔAzimuth	ΔElevation
Height bias (this work)	∼0, random	Uniform positive
Azimuthal misalignment	Systematic trend	Correlated with Az
RTK-GPS bias	Random	Random
Mirror alignment	Per-telescope	Per-telescope

The MD data unambiguously match the height-bias signature.

5 Corrected Coordinates

5.1 Original Markers: ITRF00 Values

The corrected coordinates are taken directly from the ITRF00 columns of the original OPUS reports. Where multiple solutions exist, inverse-variance weighted averages are used for the ellipsoidal height; simple averages are used for the Cartesian components.

Table 5: Corrected ITRF00 coordinates for the seven original survey markers. These values should replace the NAD 83 values currently in the analysis database.

Marker	X (m)	Y (m)	Z (m)	h (m)	σ_h (m)
BR1	−1,911,712.757	−4,567,269.863	4,009,427.954	1395.053	0.004
BR2	−1,911,674.654	−4,567,248.389	4,009,469.677	1394.659	0.009
CLF2	−1,924,405.289	−4,553,771.178	4,018,597.785	1369.979	0.016
LR1	−1,943,693.195	−4,552,383.580	4,011,197.010	1537.847	0.009
LR2	−1,943,699.086	−4,552,343.485	4,011,240.156	1538.340	0.010
MD1	−1,926,374.126	−4,539,528.317	4,033,980.925	1587.222	0.004
MD2	−1,926,193.139	−4,539,634.752	4,033,940.294	1582.449	0.020

5.2 Newer Markers: Transformation from NAD 83(2011)

For markers surveyed in NAD 83(2011)(Epoch:2010.0), empirical transformation parameters were derived using two calibration points—BRSE and MD3—which have independent OPUS solutions in both NAD 83(2011) and ITRF2014/IGS08. The Cartesian offsets at epoch 2010.0 are:

$$\begin{aligned} \Delta X &= -0.781 \text{ m} \\ \Delta Y &= +1.294 \text{ m} \\ \Delta Z &= -0.059 \text{ m} \\ \Delta h &\approx -0.72 \text{ m} \end{aligned} \tag{3}$$

Cross-validation between the BRSE and MD3 calibration points shows residuals of 2–5 mm in XYZ, confirming the transformation is valid across the ~ 35 km extent of the FD triangle. These transformed positions carry larger uncertainties (~ 1–2 cm) than the directly observed ITRF positions and should be confirmed by resurvey when practical.

5.3 Markers Requiring Action

Table 6: Status of all survey markers.

Marker(s)	Status	Action	Priority
BR1, BR2, CLF2, LR1, LR2, MD1, MD2	ITRF00 values available from OPUS reports	Replace in database	**Immediate**
CLF1	OPUS solution failed	Resurvey required	**High**
BRNE, BRNW, BRSW, MD4, MDNE, MDNW, MDSE, MDSW	Approximate ITRF from NAD 83(2011) transform	Use transformed; resurvey to confirm	Medium
BR3, BRSE, MD3	Already in ITRF	No action needed	—

6 Implications for Physics Analysis

6.1 Impact on $X_{\max}$ Reconstruction

Tokuno et al. (unpublished; cited in Nakazawa 2020, slide 26) established that a 0.1° FD pointing uncertainty produces systematic uncertainties of approximately 4% in energy and 9 g/cm^2 in $X_{\max}$. The observed systematic elevation bias of $\sim$0.17° therefore implies:

- Energy systematic: $\sim$7%
- $X_{\max}$ systematic: $\sim$15 g/cm^2

For context, the mean $X_{\max}$ difference between proton and iron primaries at $10^{19.5}$ eV is approximately 100 g/cm^2 in current hadronic interaction models. A 15 g/cm^2 systematic bias is therefore significant for composition studies.

As noted by Nakazawa (2020, slide 12): "If the copter measurement is true, the $X_{\max}$ of the current analysis is underestimated." The present work identifies the root cause of the copter measurement discrepancy as a reference frame error in the survey marker coordinates, not a deficiency of the Opt-copter system or the FD optics.

6.2 Effect on Telescope-Specific Calibration

The height offset affects all telescopes at a given station equally, since they all reference the same survey markers. This means:

- The relative telescope-to-telescope geometry within each station is unaffected.
- Total station measurements anchored to the biased markers shift rigidly.
- The absolute pointing of the entire station shifts in elevation.
- Stereo reconstruction using two stations (e.g., BRM × LR) is affected because each station has a slightly different height bias.

7 Recommendations

1. **Immediate:** Replace the NAD 83 ellipsoidal heights in the analysis database with the ITRF00 values from Table 5 for all seven original markers.

2. **Short-term:** Resurvey CLF1 using a dual-frequency GNSS receiver (e.g., SparkFun RTK Facet) with EarthScope NTRIP corrections in ITRF.
3. **Short-term:** Verify the transformed NAD 83(2011) marker positions (BRNE, BRNW, BRSW, MD4, MDNE, MDNW, MDSE, MDSW) with brief RTK occupations.
4. **Medium-term:** Reprocess the Opt-copter direction analysis at all three FD stations using corrected coordinates to confirm that the systematic elevation bias is resolved.
5. **Medium-term:** Evaluate the impact on published $X_{\max}$ and energy spectrum results by rerunning the shower reconstruction with corrected FD geometry.
6. **Long-term:** Bring all marker coordinates into a single ITRF2014 realization at a common epoch using the NGS Horizontal Time-Dependent Positioning (HTDP) utility, ensuring full consistency for future analyses.

8 Conclusion

A systematic 0.73 m vertical bias in the fluorescence detector survey marker coordinates has been identified, traced to the use of NAD 83 ellipsoidal heights where the Opt-copter analysis requires ITRF/WGS 84 values. The corrected ITRF coordinates exist in the original 2011 OPUS reports and require no new observations for the seven primary markers. The predicted elevation angle bias of 0.167° at 250 m flight distance matches the +0.17° mean offset independently measured by Matsuzawa (2025) across all 14 Middle Drum telescopes, providing strong confirmation that this reference frame discrepancy is the dominant source of the systematic elevation error observed in the Opt-copter direction analysis over the past seven years of measurements.

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Data Availability. The complete OPUS solution reports, the multi-solution averaging analysis, and the ITRF conversion spreadsheet (`TA_Markers_ITRF_Conversion.xlsx`) are available from the author. All original OPUS datasheets are archived and can be resubmitted to OPUS for independent verification.